Salem Lahlou

Assistant Professor of Machine Learning

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Research interests : AI for science tools, generative flow networks, LLM reasoning.

Experience

Sep 2024 – **Assistant Professor of Machine Learning**, *MBZUAI*, Abu Dhabi.
2024 **Senior Researcher**, *Technology Innovation Institute*, Abu Dhabi.
2018 – 2023 **Researcher**, *Mila*, Montréal - Canada.
2017 **Data Scientist**, *Booking.com*, Amsterdam.

Internships

2020 **Researcher**, *Google Research*, Paris.
2015 **Researcher**, *IBM Research*, Singapore.
2014 **Data Scientist**, *Infosys*, Bangalore.

Education

2018 – 2023 **Mila — Université de Montréal**, *Montréal - Canada*, PhD, Supervisor : Yoshua Bengio.
2015 – 2016 **École Normale Supérieure**, *Cachan - France*, M.Sc..
2012 – 2016 **École Polytechnique**, *Palaiseau - France*, B.Sc. + M.Sc..

Research grants (awarded)

Jul 2026 **IAAI Research Program grant : "Trustworthy Agronomic Reasoning for Smallholder Advisory" (24 months)**, *PI ; Co-PIs : Martin Takáč, Nils Lukas*, 500,000 USD over 24 months.
Apr 2026 **Grant from MBZUAI for research on advancing GFlowNet research**, *PI, Co-PI : Nils Lukas*, 500,000 AED.
Jun 2025 **Etihad flagship research project (24 months)**, *Co-PI, PI : Nils Lukas*, 450,000 USD per year.
Jan 2025 **EnergyAI flagship research project (30 months)**, *Co-PI, PI : Martin Takáč*, 2M USD per year.

Selected publications

(* : equal contribution)

2026 **PROBE : Learning to Audit Policy Compliance in Tool-Using LLM Agents**, *Kshitij Mishra, Abhijith Sharma, Nils Lukas,* ***Salem Lahlou****. NeurIPS 2026.*

2026 **Expert-Aided Causal Discovery of Ancestral Graphs**, *Tiago da Silva, Bruna Bazaluk, Eliezer de Souza da Silva, António Góis,* ***Salem Lahlou****, Dominik Heider, Samuel Kaski, Diego Mesquita, Adèle Helena Ribeiro. Information Sciences.*

2026 **Particle GFlowNets : Rethinking Generative Marginalization Models**, *Tiago Silva, Diego Mesquita,* ***Salem Lahlou****. UAI 2026.*

2026 **Path-dependent Discrete Amortized Inference (Oral)**, *Tiago Silva, Nikolay Malkin,* ***Salem Lahlou****. ICML 2026.*

2026 **Noise-Corrected GRPO : From Noisy Rewards to Unbiased Gradients**, *Omar El Mansouri, Fathinah Asma Izzati, Mohamed El Amine Seddik,* ***Salem Lahlou****. ICML 2026.*

2026 **CoRe : Collaborative Reasoning via Cross Teaching**, *Kshitij Mishra, Mirat Aubakirov, Martin Takáč, Nils Lukas,* ***Salem Lahlou****. ICML 2026.*

2026 **FinChain : A Symbolic Benchmark for Verifiable Chain-of-Thought Financial Reasoning**, *Zhuohan Xie, Dhruv Sahnan, Debopriyo Banerjee, Georgi Georgiev, Rushil Thareja, Hachem Madmoun, Jinyan Su, Aaryamonvikram Singh, Yuxia Wang, Rui Xing, Fajri Koto, Haonan Li, Ivan Koychev, Tanmoy Chakraborty,* ***Salem Lahlou****, Veselin Stoyanov, Preslav Nakov. ACL 2026.*

2026 **Temporal Contrastive Decoding : A Training-Free Method for Large Audio-Language Models**, *Yanda Li, Yuhan Liu, Zirui Song, Yunchao Wei, Martin Takáč,* ***Salem Lahlou****. Findings of ACL 2026.*

2026 **Byzantine Cheap Talk : Adversarial Resilience and Topology Effects in LLM Coordination Games**, *Aya El Mir, Martin Takáč,* ***Salem Lahlou****. NETYS 2026.*

2026 **Communication Enables Cooperation in LLM Agents : A Comparison with Curriculum-Based Approaches**, *Hachem Madmoun,* ***Salem Lahlou****. EACL 2026 (Oral).*

2026 **TorchGFN : A PyTorch GFlowNet Library**, *Joseph D. Viviano*, Omar G. Younis*, Sanghyeok Choi*, Victor Schmidt, Yoshua Bengio,* ***Salem Lahlou****. JMLR 2026(+ NeurIPS 2025 Workshop on Frontiers in Probabilistic Inference).*

2025 **Loss-guided Auxiliary Agents for Overcoming Mode Collapse in GFlowNets**, *Idriss Malek, Aya Laajil, Abhijith Sharma, Eric Moulines,* ***Salem Lahlou****. AAAI 2026.*

2025 **Improved Exploration in GFlowNets via Enhanced Epistemic Neural Networks**, *Sajan Muhammad,* ***Salem Lahlou****. ICoIAS 2025.*

2025 **PORT : Preference Optimization on Reasoning Traces**, ***Salem Lahlou****, *Abdalgader Abubaker*, Hakim Hacid. NAACL 2025.*

2025 **On the Privacy Risks of Spiking Neural Networks : A Membership Inference Analysis**, *Junyi Guan*, Abhijith Sharma*, Chong Tian,* ***Salem Lahlou****. UAI 2025 (+ oral at ICLR 2025 Workshop ICBINB).*

2025 **SVRPBench : A Realistic Benchmark for Stochastic Vehicle Routing Problem**, *Ahmed Heakl*, Yahia Salaheldin Shaaban*, Martin Takac,* ***Salem Lahlou****, Zangir Iklassov. NeurIPS 2025.*

2023 **A Theory of Continuous Generative Flow Networks**, ***Salem Lahlou****, Tristan Deleu, Pablo Lemos, Dinghuai Zhang, Alexandra Volokhova, Alex Hernández-García, Léna Néhale Ezzine, Yoshua Bengio, Nikolay Malkin. ICML 2023.*

2023 **GFlowNet Foundations**, *Yoshua Bengio*,* ***Salem Lahlou****, Tristan Deleu*, Edward J. Hu, Mo Tiwari, Emmanuel Bengio. JMLR 2023.*

2023 **GFlowNets and Variational Inference**, *Nikolay Malkin*,* ***Salem Lahlou****, Tristan Deleu*, Xu Ji, Edward Hu, Katie Everett, Dinghuai Zhang, Yoshua Bengio. ICLR 2023.*

2023 **DEUP : Direct Epistemic Uncertainty Prediction**, ***Salem Lahlou****, *Moksh Jain*, Hadi Nekoei, Victor Butoi, Paul Bertin, Jarrid Rector-Brooks, Maksym Korablyov, Yoshua Bengio. TMLR 2023.*

2019 **BabyAI : A Platform to Study the Sample Efficiency of Grounded Language Learning**, *Maxime Chevalier-Boisvert*, Dzmitry Bahdanau*,* ***Salem Lahlou****, Lucas Willems, Chitwan Saharia, Thien Huu Nguyen, Yoshua Bengio. ICLR 2019.*

Service (selected)

2026–2027 **ML Program Admissions Lead**, *Graduate (MSc and PhD) admissions for the Machine Learning program, MBZUAI CMS division.*

2026–2027 **Area Chair**, *ICLR 2026 and 2027, ICML 2026, NeurIPS 2026, ICOMP 2026, AAAI 2027 (Senior PC).*

2026 **Local Chair**, *AISTATS 2026.*

2025–2026 **Program Chair / PC Co-Chair**, *NETYS 2025 ; WebConf 2026 PhD Symposium ; FAISys 2026 (Xiamen, December).*

2026 **Main Organizer**, *MBZUAI ML Winter School 2026.*

2022–2026 **Reviewing**, *ICLR, ICML, NeurIPS, UAI, AAAI, ECAI, TMLR, IEEE TPAMI, JASA, ACL, NETYS, and more.*

2025–2026 **MBZUAI committees**, *Faculty Hiring, Admissions (ML + Undergraduate), ML Curriculum redesign (Jan–May 2025), TA Chair (Jan–Oct 2025).*

2025–2026 **Jury member**, *1 PhD Defense jury ; 2 PhD Candidacy juries ; 12 MSc Thesis juries.*